MIL-C-81/6C(USAF)
2 February 1976
SUPERSEDING
MIL-C-81/6B(USAF)
20 June 1975

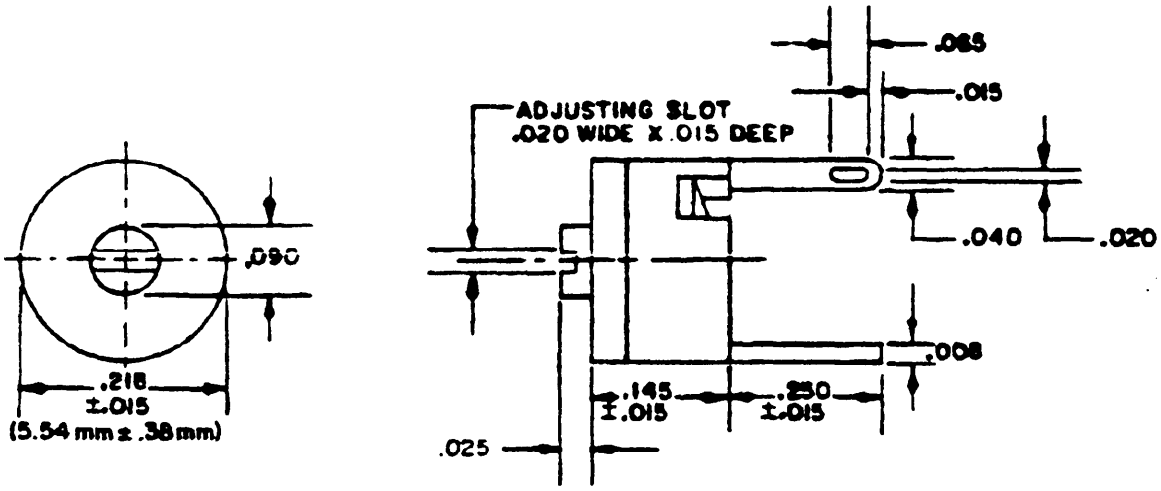
MILITARY SPECIFICATION SHEET

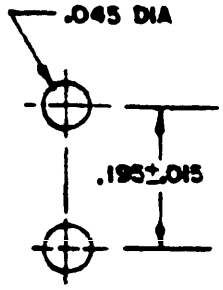
CAPACITORS, VARIABLE, CERAMIC DIELECTRIC,

STYLES CV35 AND CV36

This specification is approved for use by Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-81.





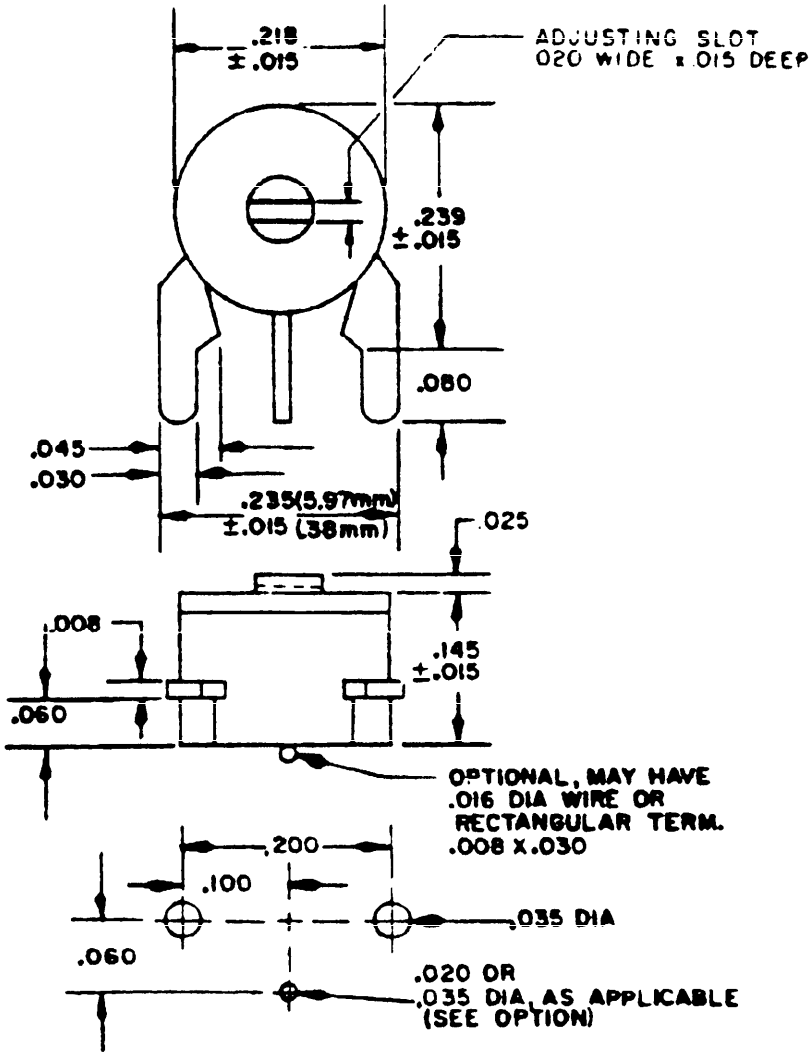
RECOMMENDED MOUNTING HOLES

Style CV35

© denotes changes.

FSC 5910

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RECOMMENDED MOUNTING HOLES

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ± .005.

Style CV 36

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REQUIREMENTS:

Dimensions and configuration: See figure.

Capacitance value: See table I.

DC rated voltage: See table I.

Operating temperature range:
Characteristic A - -55° to +85° C; linearly derate to 50 Vdc for +125° C operation.
Characteristic G - -55° to +125° C.

Characteristic: See table I.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
A dc potential of 200 Vdc applied between terminals for 3 ±2 seconds.

Barometric pressure (reduced):
Method 105 of MIL-STD-202, condition D (100,000 ft).
Test potential: 100 percent of dc rated voltage.

Insulation resistance:
Method 302 of MIL-STD-202, condition A, 100 volts dc applied - 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

© DF: At 1 MHz ±100 kHz, at maximum and minimum capacitance - Shall be not more than 0.2 percent for characteristic A and 0.5 percent for characteristic G.

TABLE I. Styles CV35 and CV36.

Type 1/ designation	Capacitance (pF)		DC rated voltage (volts)	Characteristic						
					Capacitance change from value at 25° C					
					At -55° C		At +85° C		At +125° C	
	Minimum	Maximum		Symbol	Minimum percent	Maximum percent	Minimum percent	Maximum percent	Minimum percent	Maximum percent
CV35A030	1.0	3.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
CV36A030	1.25	3.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
CV3-A0502/	2.0	5.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
CV3-A090	2.5	9.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
CV3-A200	3.5	20.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
CV3-A1502/	5.0	15.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
CV3-A250	5.0	25.0	100	A	-4.5	+2.0	-2.5	+2.0	-4.2	+3.4
© CV3-G400	7.0	40.0	25	G	0.0	+14.0	-8.0	-3.0	-14.0	-5.0

1/ Where applicable, the complete type designation will include an additional digit to indicate CV35 or CV36.

2/ Characteristic F is hereby canceled and there is no superseding part.

Temperature coefficient:
Within the limits specified for the applicable characteristic.

Capacitance drift:
Within 0.75 percent of initial step 1 measurement or 0.50 picofarad (pF), whichever is greater.

Terminal strength:
Pull test - Capacitor held by body and 4-pound load applied to each terminal for at least 10 seconds.

Torque:
Not less than .3 nor more than 3 ounce-inches.

Shock (specified pulse):
Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency:
Method 204 of MIL-STD-202, condition B (15G).
Capacitance change - Shall not exceed ±2 percent or 0.5 pF, whichever is greater.
© DF - Shall be not more than 0.2 percent for characteristic A and 0.5 percent for characteristic G.
Dielectric withstanding voltage:
200 volts dc, applied for 3 ±2 seconds.
Insulation resistance:
10,000 megohms, minimum.

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Fatigue:
Capacitance change - Shall not exceed ± 12 percent of initial value or 0.75 pF, whichever is greater.
Torque - Not less than .3 nor more than 3 ounce-inches.

Life.
Qualification test - 1,000 hours at + 85 °C, 150 percent of rated volts dc with a peak alternating voltage of 50 percent of rated volts dc (100 Hertz or less) superimposed.
Insulation resistance - Initial requirement.
Capacitance change - Shall not exceed ± 8 percent of initial value or 0.5 pF, whichever is greater.

Group C life: Conditions and requirements are the same as that for qualification except that capacitance change shall not exceed ± 5 percent of initial value or 0.5 pF, whichever is greater.

Moisture resistance:
Method 106 of MIL-STD-202:
Insulation resistance - 10,000 megohms, minimum.
Capacitance change - Shall not exceed ±5 percent of nominal value or 0.5 pF, whichever is greater.
© DF - Shall be not more than 0.5 percent for characteristic A and 1.25 percent of the initial requirement for characteristic G.

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-F284)

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